Learning one magnetic model for all winding connections: connection-invariant co-energy surrogates and drive-cycle co-design of reconfigurable electric machines

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Abstract

Reconfigurable windings (series–parallel, star–delta, taps, open-end) extend the speed range and part-load efficiency of electric machines, yet every candidate connection is still evaluated with its own dedicated finite-element (FE) model or flux-map table, leaving the combinatorial connection space unexplorable at design time. We treat a winding connection as what it physically is: linear constraints on the coil currents, $\boldsymbol{i}_c = \boldsymbol{C}\boldsymbol{i}_p$. Because the magnetic co-energy is connection-invariant, a single *connection-invariant surrogate*, trained once on coil-level FE data, evaluates any connection zero-shot. We formalise this generalisation problem with an error bound governed by the coverage of the training current distribution (rank correlation 0.82 over seven hold-outs); instantiate the surrogate with a structured co-energy network whose reciprocity and torque–flux consistency hold by construction (4.6×10^{-4} vs 0.22 for a black-box MLP); and release a seven-connection, three-tier benchmark on a 12-slot/4-pole machine. The benchmark yields two data-collection principles: what *kind* of current patterns is covered dominates how many samples exist. Magnitude- random patterns (48 groups) reduce an unbalanced hold-out from flux nRMSE 1.00 to 0.11 but leave an open-phase connection at 1.38; 24 *phase*-random groups—a quarter of the FE cost—bring it to 0.07 and the star–delta hold-out from 0.27 to 0.07; both principles replicate on a second machine. A monotonicity-constrained architecture fails outright (≥ 0.28 everywhere): the joint co-energy is not convex. Finally, on the official WLTC class-1 cycle, a dynamic-programming scheduler

reaches full torque tracking at 440.4 Wh, essentially equal to the lowest energy obtained by the nearly complete fixed delta connection (440.6 Wh) while removing its tracking deficit; the 15.1 % advantage over the only fully feasible fixed connection mainly reflects that connection's low-speed current-class penalty. An independent FEMM cross-check reproduces the linear inductance structure and nonlinear flux trends, but also exposes a failed zero-shot torque screen for the untrained tap connection; we therefore separate validated modelling claims from exploratory screening.

Keywords: winding reconfiguration, magnetic co-energy, surrogate modelling, zero-shot generalisation, physics-informed neural networks, drive-cycle optimisation

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1. Introduction

1.1. Motivation

Reconfigurable machine windings—series–parallel changeover, star–delta switching, mid-winding taps, open-end windings and post-fault reconfiguration—trade voltage class against current class at run time. Recent reviews [1, 23] report torque and power-density gains of 40–80 %, machine loss reductions of 30–60 %, and inverter current reductions above 40 % on a production passenger car [1]. The enabling physics has been demonstrated repeatedly: series connection multiplies the back-EMF and suits low-speed high-torque operation, while parallel connection divides it and extends the constant-power speed range [2, 3].

What has *not* kept pace is the modelling machinery. Every published study evaluates two to four hand-picked connections, each with its own FE model or its own dq flux-map table. For a machine with four coil groups per phase the discrete connection space already contains hundreds of meaningful entries (series–parallel groupings, taps, star/delta, open-winding, degraded states), and each FE campaign takes hours to days. The 2025 review is explicit: the field lacks (i) early-stage design methods that compare connections systematically, (ii) drive-cycle-level switching strategies, and (iii) system-level loss accounting

that includes the changeover circuit [1]. Concurrently, the transient-control side of the problem has been pushed to near-optimal territory—recent deadbeat predictive control achieves 5 μs seamless changeover with no torque interruption [4, 5, 6]—so the bottleneck has decisively shifted from "how to switch" to "what to switch to, and when".

1.2. Related work and the gap

Analytic connection selection.. Normalised characteristic-current criteria map connection families onto a single $\hat{I}_{ch}$–ξ plane and give closed-form decision thresholds [7]. These are valuable screening tools, but they are derived under the assumptions of linearity, negligible cross-saturation and negligible losses; the same paper lists drive-cycle optimisation and switching transients as open problems. Analytic criteria cannot exploit the nonlinear multi-coil physics that FE already provides.

Learned magnetic surrogates.. Neural mappings of flux linkage and torque are standard for fixed windings: ANN torque surrogates [8], physics-informed networks [18] including PINN-parameterised PMSM fields [9], POD/SVR fast surrogates [10], Fourier neural operators [19] for FE acceleration [11], and data-efficient *gradient networks* that learn the magnetic co-energy/flux-linkage map with reciprocity and energy conservation enforced by the architecture [12]. All of these operate on the d–q plane of *one* fixed winding; changing the connection changes the input dimension itself, so the model must be retrained. None of them addresses topology generalisation.

Winding-level reconfiguration studies.. Series–parallel changeover hardware and control are mature [2, 13, 14]; open-end and multi-mode drives select the best mode pointwise [15]; three-speed tapped IPMSMs [16] and reconfigurable SRM windings [17] extend the principle. Mode selection, however, is either a fixed speed threshold or a per-point greedy choice; no study optimises the *sequence* of modes over a drive cycle with changeover losses and dwell constraints, and losses of the changeover circuit itself are usually ignored [1].

The gap.. Three communities hold one piece each: connection physics (electrical machines), structure-enforcing ML (co-energy/gradient networks), and scheduling (energy management). What is missing is the bridge: *a single magnetic model that any connection can query.* If the connection is only a linear constraint on coil currents, such a model turns connection exploration from "one FE campaign per connection" into "one training run plus a matrix product", and makes drive-cycle co-design of topology and schedule computationally trivial.

1.3. Contributions

1. **Problem formulation and theory (§2).** We cast winding reconfiguration as constrained regression in coil-current space, prove that the co-energy and its derivatives are connection-invariant (Prop. 1), and give an error bound for zero-shot transfer to unseen connections that is governed by the coverage distance of the training current distribution (Prop. 2). The bound predicts what the experiments show: connections whose current vectors lie inside the training support transfer at in-distribution accuracy, while support-extrapolating connections fail until the support is extended; the coverage distance correlates with zero-shot error at rank $\rho = 0.82$ over seven hold-outs, with informative misalignments that delimit when the screen applies.
2. **Connection-invariant structured co-energy network (§3).** A scalar co-energy head with a Fourier-parameterised, PSD-seeded inductance matrix, harmonic PM flux and a zero-initialised residual MLP. Reciprocity, energy conservation and torque–flux consistency hold by construction to autograd precision (4.6×10^{-4} measured), versus 0.22 for a same-size MLP. The structure is offered as a consistency option, not an accuracy winner: an unstructured MLP remains more accurate on in-support hold-outs and on the torque channel, and the benchmark quantifies this trade-off rather than claiming dominance. We also report an instructive negative result: a monotone (PSD-Jacobian, input-convex-style [20]) vector-field parameterisation fails at all difficulties because the joint co-energy is not convex

in $(\boldsymbol{i}_c, \theta)$.

3. **A stratified zero-shot benchmark with data-design principles (§4–5).** Seven hold-out connections across three difficulty tiers, 9,276 nonlinear FE samples (23,000+ including the drive-cycle pools), four model families, and a reproducible split protocol. Two findings generalise beyond this machine: (i) 48 magnitude-random current patterns rescue unbalanced connections (nRMSE $1.00 \to 0.11$) where doubling standard-connection data does not help; (ii) phase-random patterns (a quarter of the magnitude family's FE cost) rescue open-phase connections that magnitude patterns cannot ($1.38 \to 0.07$). Coverage *kind* dominates coverage *amount*; both principles replicate on a 6-slot/4-pole second machine.
4. **Drive-cycle co-design and zero-shot screening (§6).** The surrogate generates mode-conditioned loss maps in milliseconds; a dynamic program over the official WLTC class-1 trace jointly selects connection and operating point under voltage envelopes, current limits, minimum dwell and changeover energy. The resulting schedule completes the cycle with essentially the same energy as the best nearly feasible fixed connection, while the nominal 15.1 % saving against the only fully feasible fixed connection is reported together with that baseline's low-speed current-class penalty. Loss decomposition and a changeover-energy sweep spanning four orders of magnitude show that this conclusion is robust to the switching model. A separate FEMM cross-check is used to delimit rather than overstate zero-shot screening: an untrained tap connection preserves the coarse flux trend but not torque along a synchronous trajectory, so we remove the earlier unsupported claim that this mode is dynamically dominated.

Data, code, trained models and the benchmark protocol are released; all numbers in this paper are reproducible with one command per figure.

2. Problem formulation

2.1. Notation

Consider a machine with N_c independently excitable slot coils (here $N_c = 12$), rotor position θ, coil currents $\boldsymbol{i}_c \in \mathbb{R}^{N_c}$, and phase/port currents $\boldsymbol{i}_p \in \mathbb{R}^{N_p}$. A *winding connection* is a linear map

$$\boldsymbol{i}_c \;=\; \boldsymbol{C}\,\boldsymbol{i}_p, \qquad \boldsymbol{C} \in \mathbb{R}^{N_c \times N_p}, \tag{1}$$

whose entries encode turns, polarity and group membership. Series–parallel groupings, star/delta (with delta line currents absorbed into $\boldsymbol{C}$), taps (zero columns for idle coils), open-end windings and open-phase degradation are all instances of (1). The seven connections used in this paper, together with the structure of their $\boldsymbol{C}$, are as follows: 4S (series, $\boldsymbol{C}$ one column of ± 1 per phase), 2S2P and 4P (2 and 4 parallel branches, coil currents scaled by 1/2 and 1/4), delta (winding currents from line currents through the standard 1/3-transform), asym (3+1/2+2/1+3 series–parallel splits with inverse-resistance branch sharing), tap50 (half the coils idle: zero columns), and openB (phase-B column removed, isolated-neutral constraint $i_A + i_C = 0$ folded into the remaining columns).

2.2. Connection invariance of the co-energy

Proposition 1 (Connection invariance)**.** *Fix the geometry and materials. Let $W'(\theta, \boldsymbol{i}_c)$ be the magnetic co-energy of the nonlinear magnetostatic problem. Then for every connection $\boldsymbol{C}$ and every admissible port current $\boldsymbol{i}_p$ with $\boldsymbol{i}_c = \boldsymbol{C}\boldsymbol{i}_p$:*

$$\boldsymbol{\psi}_c = \frac{\partial W'}{\partial \boldsymbol{i}_c}, \qquad T = \frac{\partial W'}{\partial \theta}, \qquad \boldsymbol{\psi}_p = \boldsymbol{C}^{\top}\boldsymbol{\psi}_c, \tag{2}$$

and the triple $(\boldsymbol{\psi}_c, T, W')$ is independent of $\boldsymbol{C}$.

Proof. The magnetostatic problem is posed on coil currents only; $\boldsymbol{C}$ enters exclusely through (1), so W' is evaluated at the same point of the same function regardless of how that point is parametrised by ports. The port flux follows from power invariance: the magnetic power at the ports equals the co-energy rate,

$\boldsymbol{\psi}_p^\top \dot{\boldsymbol{i}}_p = \boldsymbol{\psi}_c^\top \dot{\boldsymbol{i}}_c = \boldsymbol{\psi}_c^\top \boldsymbol{C} \dot{\boldsymbol{i}}_p$ for all $\dot{\boldsymbol{i}}_p$, hence $\boldsymbol{\psi}_p = \boldsymbol{C}^\top \boldsymbol{\psi}_c$. Torque is the θ-partial of the same scalar. (Numerical verification on the FE solver: reciprocity of the differential inductance matrix to 6.4×10^{-4} and path independence of $\int \boldsymbol{\psi}_c \cdot d\boldsymbol{i}_c$ to 0.62 %, Table 1.) □

Proposition 1 is elementary, but its computational consequence is the point of this paper: *a surrogate of $(\theta, \boldsymbol{i}_c) \mapsto (\boldsymbol{\psi}_c, T)$ trained once is a surrogate for every connection simultaneously.* Evaluating a new connection costs a matrix product and the solution of a small port-level circuit model, not a new FE campaign.

2.3. What zero-shot transfer costs

Training data live on a union of subspaces $\mathcal{S} = \{\boldsymbol{i}_c = \boldsymbol{C}_k \boldsymbol{i}_p : \boldsymbol{i}_p \in \mathcal{I}_k\}_{k=1}^K$ determined by the K training connections and their current ranges $\mathcal{I}_k$. A hold-out connection $\boldsymbol{C}_*$ queries the model along a different subspace $\mathcal{S}_*$. Let $d(\boldsymbol{i}_c, \mathcal{S}) = \min_{\boldsymbol{s} \in \mathcal{S}} \|\boldsymbol{i}_c - \boldsymbol{s}\|$ and let L_ψ be a Lipschitz constant of the true flux map.

Proposition 2 (Coverage bound)**.** *Suppose $\|\hat{\boldsymbol{\psi}}_c(\theta, \boldsymbol{i}_c) - \boldsymbol{\psi}_c(\theta, \boldsymbol{i}_c)\| \le \varepsilon$ for all $(\theta, \boldsymbol{i}_c)$ with $\boldsymbol{i}_c \in \mathcal{S}$ (training-domain accuracy), and both $\hat{\boldsymbol{\psi}}_c$ and $\boldsymbol{\psi}_c$ are L-Lipschitz in $\boldsymbol{i}_c$ on a convex hull containing $\mathcal{S} \cup \mathcal{S}_*$. Then for every $\boldsymbol{i}_{c,*} \in \mathcal{S}_*$,*

$$\|\hat{\boldsymbol{\psi}}_c(\theta, \boldsymbol{i}_{c,*}) - \boldsymbol{\psi}_c(\theta, \boldsymbol{i}_{c,*})\| \;\le\; \varepsilon + (L + L_{\hat{\psi}})\, d(\boldsymbol{i}_{c,*}, \mathcal{S}).$$

The same statement holds port-wise with an extra factor $\|\boldsymbol{C}_*\|_2$ for $\boldsymbol{\psi}_p = \boldsymbol{C}_*^\top \boldsymbol{\psi}_c$.

Proof. Pick $\boldsymbol{s} \in \mathcal{S}$ attaining the distance. Two triangle inequalities give $\|\hat{\boldsymbol{\psi}}_c(\boldsymbol{i}_{c,*}) - \boldsymbol{\psi}_c(\boldsymbol{i}_{c,*})\| \le \|\hat{\boldsymbol{\psi}}_c(\boldsymbol{i}_{c,*}) - \hat{\boldsymbol{\psi}}_c(\boldsymbol{s})\| + \|\hat{\boldsymbol{\psi}}_c(\boldsymbol{s}) - \boldsymbol{\psi}_c(\boldsymbol{s})\| + \|\boldsymbol{\psi}_c(\boldsymbol{s}) - \boldsymbol{\psi}_c(\boldsymbol{i}_{c,*})\| \le L_{\hat{\psi}} d + \varepsilon + L d$. □

The bound is qualitative: neither L nor $L_{\hat{\psi}}$ is estimated, so it predicts ordering tendencies, not numeric error. Its value is the screening logic—distance to the support is the only channel through which a connection can hurt—which §5 evaluates empirically with the measured $\bar{d}$.

Remark 1. *The bound contains no term for the* label *of the connection—only the geometric distance of its current vectors to the training support. It is consistent with the experimental stratification of §5: scaled copies of a training subspace (4P of 4S) transfer perfectly; phase-shifted combinations (delta) transfer partially; unbalanced current ratios (asym) transfer only after magnitude-random patterns shrink d; and open-phase patterns, which contain current vectors with sign-flipped phase pairs that magnitude perturbations never produce, require* phase-*random patterns. Section 5 computes $\bar{d}$ per hold-out and reports the rank correlation. The bound also explains why more of the wrong data does not help: d is a property of the support geometry, not of the sample density.*

2.4. Learning problem

Given samples $\{(\theta_j, \boldsymbol{i}_c^{(j)}, \boldsymbol{\psi}_c^{(j)}, T^{(j)})\}_{j=1}^{N}$ on $\mathcal{S}$, learn $\hat{\boldsymbol{\psi}}_c, \hat{T}$ (or a potential $\hat{W}'$) minimising held-out error on $\mathcal{S}$, and evaluate *zero-shot* on subspaces $\mathcal{S}_* \not\subseteq \mathcal{S}$. Model selection uses only in-support validation data; hold-out connections are touched once, at reporting time. This protocol is stricter than the common practice of early-stopping on the test distribution and is part of the released benchmark.

3. Connection-invariant surrogates

All models share the input $x = [\boldsymbol{i}_c/60,\ \phi(\theta)] \in \mathbb{R}^{19}$ with harmonic features $\phi = [1, \sin 2\theta, \cos 2\theta, \sin 4\theta, \cos 4\theta, \sin 6\theta, \cos 6\theta]$ (electrical harmonics of the 4-pole machine), and predict the 13 outputs $(\boldsymbol{\psi}_c, T)$, standardised by training statistics.

3.1. Structured co-energy network (ours)

The co-energy head is a scalar field

$$\hat{W}'(\boldsymbol{i}_c, \phi) = \tfrac{1}{2}\, \boldsymbol{i}_c^{\top} \mathbf{L}(\phi)\, \boldsymbol{i}_c + \boldsymbol{p}(\phi)^{\top} \boldsymbol{i}_c + R_\theta(\boldsymbol{i}_c, \phi), \tag{3}$$

with the three terms parameterised as follows.

Inductance matrix. $\mathbf{L}(\phi) = \mathbf{A}\mathbf{A}^\top + \epsilon\mathbf{I} + \sum_{h\in\{4,8\}} \left[\mathbf{C}_h \cos h\theta + \mathbf{S}_h \sin h\theta\right]$, each $\mathbf{C}_h, \mathbf{S}_h$ symmetrised. The $\mathbf{A}\mathbf{A}^\top + \epsilon\mathbf{I}$ seed makes the current-independent part positive definite, so at initialisation the model reduces to a stable linear-inductance model; the angle harmonics (orders 4 and 8, i.e. the slotting harmonics of the 12-slot/4-pole geometry) modulate it. Cos/sin of 8θ are composed exactly from the ϕ columns via double-angle identities, keeping a single autograd path.

PM flux. $\boldsymbol{p}(\phi) = \mathbf{P}\phi$, a full harmonic PM flux linkage per coil (orders 2, 4, 6 plus offset).

Saturation residual. R_θ is an MLP on $(\boldsymbol{i}_c, \phi)$ whose final layer is *zero-initialised*. Training therefore starts from the closed-form linear+PM model and grows a correction only where the data demand it—the same spirit as physics-model-plus-residual learning, but with the residual inside a scalar potential.

Flux and torque are obtained by automatic differentiation,

$$\hat{\boldsymbol{\psi}}_c = \nabla_{\boldsymbol{i}_c} \hat{W}', \qquad \hat{T} = \nabla_\phi \hat{W}' \cdot \frac{d\phi}{d\theta}, \tag{4}$$

so both outputs are exact gradients of one scalar. The training loss up-weights the single torque channel ($w_T = 3$) to compensate for its 1/13 gradient share.

Proposition 3 (Structural consistency)**.** *For (3)–(4), (i) the Jacobian $\partial\hat{\boldsymbol{\psi}}_c/\partial\boldsymbol{i}_c = \mathbf{L}(\phi) + \nabla^2_{\boldsymbol{i}_c} R_\theta$ is symmetric (reciprocity) exactly; (ii) the mixed identity $\partial\hat{\psi}_k/\partial\theta = \partial\hat{T}/\partial i_k$ holds exactly (torque–flux consistency); (iii) $\hat{W}'$ is a well-defined scalar potential, so no operating point can exhibit non-conservative flux–torque behaviour in downstream optimisation.*

Proof. (i) is the symmetry of the Hessian of a twice-differentiable scalar; (ii) is the commutation of mixed partials; (iii) is immediate. □

3.2. Baselines

Direct MLP. A 3-layer width-128 GELU network regressing the 13 outputs: the accuracy ceiling without structure. **Per-connection MLP.** The industrial status quo: a direct MLP trained only on the nominal connection's data (4S),

reported on hold-out connections. **Monotone vector field (negative result).** The architecture of [12] generalised to 12 coils: $g(x) = \mathbf{A}_0 x + \boldsymbol{b}_0 + \mathbf{A}_1^\top \sigma(\mathbf{A}_1 x + \boldsymbol{c})$ with σ = softplus and $\mathbf{A}_0 = \mathrm{blkdiag}(\mathbf{B}^\top\mathbf{B} + \epsilon\mathbf{I}, 0)$, which makes the field g the exact gradient of a convex scalar and its Jacobian symmetric positive semidefinite everywhere. Despite perfect reciprocity (1.6×10^{-5}, Table 5), it cannot fit this machine at any difficulty tier (§5.4). **Reciprocity-structured ridge.** Linear-in-current model $\hat{\psi}_i = \sum_h H_h(\theta)\big[a_{ih} + \sum_{j\le k} S_{ijk}\, m_{jk}(\boldsymbol{i}_c)\big]$ with symmetric S—the "linear physics" reference.

3.3. Training protocol

Full-batch AdamW, 10^4–1.2×10^4 epochs, gradient clipping, and early stopping on a *stratified validation slice of the training connections only* (12 %, never on hold-out rows). Three seeds were used where results are reported. A single training run takes 3–230 s on a laptop GPU; batched inference costs tens of microseconds per point on GPU, roughly five orders of magnitude faster than the 1.2 s FE reference solve; the total compute behind every number in this paper is under three GPU-hours plus 40 CPU-core-hours of FE solving.

4. FE data and benchmark protocol

4.1. Machine and solver

The case-study machine is a 12-slot/4-pole surface-mounted PM machine, solved with an in-house 2-D nonlinear magnetostatic solver: Fröhlich-type iron $\mu_r(B)$, radial PMs, one independent current per slot, coil flux linkage from the vector potential, and Lorentz torque on the slot conductors (cogging is zero for this symmetric geometry; reluctance-torque effects enter through the flux linkages). The training mesh is 40×96 (≈ 1.2 s per nonlinear solve, 8-way parallel); an 80×192 re-solve of 64 probe points quantifies discretisation uncertainty (Table 1). Machine parameters: 2 pole pairs of surface PM (radial, B_r-equivalent magnetisation 1.2 T), stack length 100 mm, 40 turns per coil, slot fill 40 %, slot-pitch occupancy 0.52, Fröhlich iron ($\mu_{r0} = 4000$, $B_{sat} = 1.6$ T);

per-coil resistance 3.0 mΩ from geometry. We deliberately use a research-grade solver: the claims of this paper concern relative model behaviour against a fixed reference, not absolute machine performance.

4.2. Datasets

All datasets share the rotor grid $\theta \in \{0, \ldots, 165°\}$, step 15°. The **zero-shot benchmark pool** (9,276 samples) consists of the first, second, third and last bullets below; the phase-random family enters the coverage ablations of §5.2, and the last two bullets are generated exclusively for the drive-cycle study:

- **Standard connections** (1,344): 4S, 2S2P, delta, asym at 7×4 current-amplitude/angle combinations.
- **Magnitude-random patterns** (4,608): base 4S excitation, each coil scaled by $U(0.2, 1)$, RMS-renormalised; 48 groups.
- **Zero-support patterns** (2,304): scale $U(0, 1)$, for the tap/fault ablation.
- **Phase-random patterns** (1,152): per-phase amplitude $U(0.2, 1)$, per-phase angle $U(0, 2\pi)$, plus zero-sequence $U(-0.3, 0.3)$—current vectors with arbitrary phase relations.
- **Dense current-angle sweep** (2,304): 4S/2S2P/delta/4P over 12 current angles per 120° sector.
- **Synchronous-path set** (10,368): the four switchable modes along their actual synchronous trajectories $i_p(\theta) = I[\cos(2\theta + \gamma), \ldots]$ with γ over the full 360° electrical domain (36 values) and $I \in \{10, \ldots, 60\}$ A—the distribution the drive-cycle study queries. (The torque–current-angle characteristic of this machine has no sub-360° period: we verified with dedicated FE solves that, once saturation is active, neither the three-phase permutation symmetry nor the half-period sign symmetry holds exactly, so the full domain must be sampled.)

Table 1: FE solver consistency checks and discretisation uncertainty.

Check	Result	Interpretation
Linear mutual-inductance reciprocity	4.2×10^{-6}	solver-level
Nonlinear differential reciprocity (20 A bias)	6.4×10^{-4}	solver-level
Co-energy path independence (2 paths)	0.62 %	solver-level
Flux, 80×192 vs 40×96	ratio 1.05 ± 0.16	discretisation
Torque, 80×192 vs 40×96	median 13 %	discretisation

- **Zero-current PM set** (12) and **hold-out connections** (1,008): 4P (quarter-scaled copy of 4S), tap50 (half-winding series tap), openB (phase-B open, isolated neutral, $i_A + i_C = 0$).

4.3. Protocol

Two evaluation modes. **LOCO** (leave-one-connection-out): train on the other three standard connections plus patterns, test on the hold-out one. **EXT**: train on all four standard connections + patterns, test on 4P/tap50/openB which are never seen in any form. Three difficulty tiers follow from Prop. 2: *tier 1* in-support scaling (2S2P, 4P), *tier 2* phase-recombination (delta), *tier 3* support-extension (asym: unbalanced ratios; tap50/openB: zero/excluded coils and sign-flipped phase pairs). Metrics: flux nRMSE (per-coil RMSE normalised by per-coil test standard deviation, averaged over 12 coils) and torque R^2.

Table 2 reports an independent FEMM 4.2 cross-check of the same geometry and material curve. The linear inductance matrix has correlation 0.99993 with the reference solver; after one global unit-scale calibration its residual is 11.7 %. The PM-only waveform has correlation 0.944 and a fundamental-amplitude error of 30 % on average, while a saturated 4S case reaches flux correlation 0.989 and 17.5 % torque difference against the weighted-stress-tensor result. These discrepancies are of the same order as the fine-mesh uncertainty and preclude claims of absolute accuracy. They also reveal a more specific failure: for an untrained tap50 synchronous trajectory, the direct MLP preserves only a coarse flux trend ($r = 0.71$ after global scaling) and its torque correlation with FEMM

Table 2: Independent FEMM 4.2 cross-check. The inductance discrepancy is computed after one global scale factor; nonlinear flux is pointwise and torque uses the weighted-stress-tensor result. The tap50 rows compare the zero-shot MLP with FEMM at six points on a fixed-I synchronous trajectory.

Comparison	Correlation	Discrepancy
Linear inductance matrix (12×12)	0.99993	11.7 %
PM-only flux waveform	0.944	30.2 % fundamental
Nonlinear 4S flux (60 A)	0.989	40.9 % pointwise
Nonlinear 4S torque (60 A)	—	17.5 %
tap50 zero-shot flux	0.708	70.7 %
tap50 zero-shot torque	−0.206	not predictive

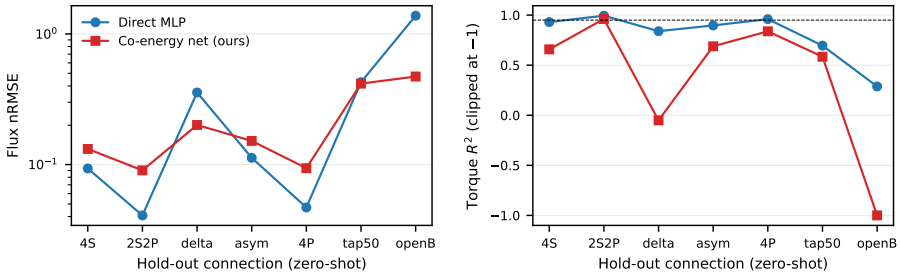


Figure 1: Zero-shot benchmark across the seven hold-out connections (base variant). Left: flux nRMSE (log scale); right: torque R^2. Tier-1 hold-outs transfer at in-distribution accuracy; tier-3 hold-outs require pattern coverage; the monotone vector field fails everywhere.

is −0.21. Zero-shot screening is therefore useful as a hypothesis generator, not as a substitute for FE verification of a design decision.

5. Zero-shot results

5.1. Main benchmark

Table 3 and Fig. 1 report all hold-outs; Fig. 2 shows parity plots for the hardest LOCO hold-out. As a quantitative anchor for Prop. 2, we compute for each hold-out the mean min-distance $\bar{d}$ of its test current vectors (normalised by the 60 A scale) to the sampled training support: 4P 0.036, 2S2P 0.055,

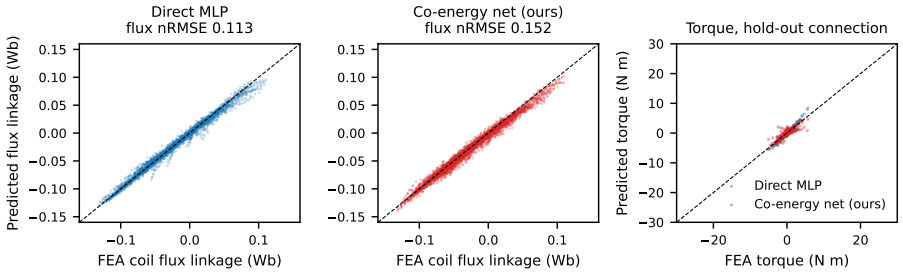


Figure 2: Hold-out connection *asym*: predicted vs FE coil flux linkages (left: direct MLP; centre: co-energy net) and torque (right).

asym 0.206, delta 0.240, 4S 0.263, openB 0.288, tap50 0.536. Across the seven hold-outs the rank correlation with the zero-shot flux error is $\rho = 0.82$ ($p = 0.02$). The correlation is meaningful but not perfect, and the misalignment is informative: openB sits at a modest $\bar{d}$ yet fails hardest (1.38), because its sign-flipped phase pairs are *close* in Euclidean distance to the training support while lying in a direction the support barely samples—an unweighted distance metric cannot see support density or direction. The 4S hold-out supplies the opposite counterexample: its distance is high (0.263) while its flux error is low (0.093). We therefore use $\bar{d}$ only as a coarse risk indicator; neither a large nor a small value justifies accepting or rejecting a connection. Four observations.

(1) In-support transfer is essentially free. For tier-1 hold-outs (2S2P, 4P) the shared model reaches flux nRMSE 0.02–0.05 and torque $R^2 \geq 0.96$, i.e. in-distribution accuracy: their current subspaces lie inside the training support ($\bar{d} < 0.06$, Prop. 2), so the connection adds no error of its own. Notably, 4P was *never seen*—it is a rescaled 4S, and the model treats it as such.

(2) Without pattern coverage, tier-3 collapses to chance. When trained on the standard connections only ("no-random"), every model predicts flux on asym at nRMSE ≈ 1.0—no better than predicting the mean. (The flux channel collapses; the torque channel partially survives at $R^2 \approx 0.95$ because the dominant average-torque trend is recoverable from the rotor-angle features alone—a caution against judging such models by torque R^2 only.) This is the

Table 3: Zero-shot benchmark, flux nRMSE / torque R^2. Training variants: "base" = three standard connections (all four for EXT rows) + 48 magnitude-random groups + PM data; "withz" = base + the 24 zero-support pattern groups; "no random" = standard connections + PM data only. LOCO rows hold out one standard connection; EXT rows (4P, tap50, openB) hold out connections never seen at all. Best per row in bold.

Tier	Hold-out	Direct MLP	Co-energy (ours)	Monotone VF
1	4S (LOCO)	**0.093**/**0.930**	0.132/0.658	0.414/−0.68
1	2S2P (LOCO)	**0.041**/**0.994**	0.090/0.961	0.382/−0.53
1	4P (EXT)	**0.047**/**0.960**	0.094/0.838	0.409/−8.2
2	delta (LOCO)	0.356/**0.840**	**0.201**/−0.05	0.424/−2.0
3	asym (LOCO)	**0.113**/**0.897**	0.152/0.689	0.495/−1.7
3	tap50 (EXT)	0.427/0.696	**0.363**/**0.811**†	0.479/−0.14
3	openB (EXT)	1.379/0.288	**0.472**/−18.5	0.563/−5.6
asym, no random patterns		1.002/0.949	1.000/0.397	0.990/−0.20
asym, per-connection MLP (4S only)		1.002/0.930	—	—
asym, linear ridge (reciprocity-structured)			0.652	

†with zero-support patterns included ("withz" variant); see §5.2.

single most important negative control in the benchmark: connection labels are worthless; current-space coverage is everything.

(3) Magnitude patterns rescue unbalance. Adding 48 magnitude-random groups brings asym to 0.11 (MLP) and 0.13–0.16 (co-energy nets); MLP torque R^2 reaches 0.85–0.90 (co-energy 0.69).

(4) Phase patterns rescue open-phase. Magnitude patterns leave openB at 1.38; 24 phase-random groups bring it to 0.066 (§5.2)—the second design principle.

5.2. What kind of coverage matters

Fig. 3 shows the coverage-scaling curves; Table 4 isolates the effect, and it is sharper than anticipated: the two pattern families rescue *disjoint* connection classes. Magnitude patterns bring tap50 from 1.86 to 0.43 (they sample non-uniform per-coil amplitudes, including small ones) but leave openB at 1.39 (open-phase operation drives sign-flipped phase pairs that multiplicative per-

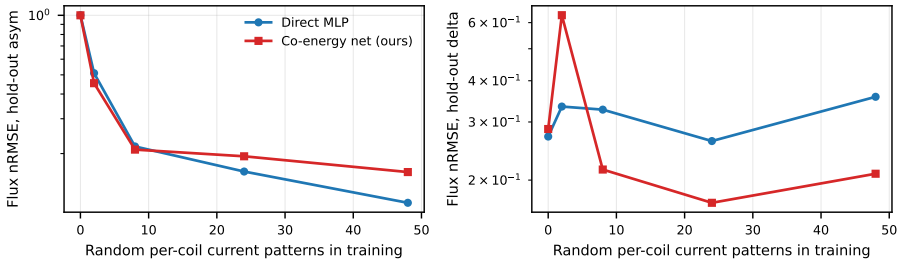


Figure 3: Coverage scaling of magnitude-random patterns on the two hardest standard hold-outs. asym improves monotonically with pattern count; delta is insensitive to it—its requirement is phase coverage (Table 4).

turbations of a balanced base never generate). Phase-random patterns do the opposite: openB drops to 0.066 ($R^2 = 0.99$) and delta from 0.27 to 0.072 (delta is itself a phase-recombination), while tap50 degrades to 2.1 in that pool alone (phase-random patterns keep all coils energised). With both families in the pool (a quarter of the magnitude family's FE cost added on top), each hold-out is served by the family it needs: openB 0.09, delta 0.08, tap50 0.43 (via either family's best). On delta, moreover, scaling the *magnitude* family from 0 to 48 groups leaves the error essentially unchanged (0.27 at zero groups, 0.26 at 24, 0.36 at 48) (0.27 → 0.36), while a quarter-sized phase family reaches 0.07: coverage *kind* dominates coverage *amount*. This is consistent with the qualitative implication of Prop. 2, although the bound itself is not tight in Euclidean distance.

5.3. Physical consistency

Table 5 and Fig. 4 evaluate the flux Jacobian $\partial\hat{\boldsymbol{\psi}}_c/\partial\boldsymbol{i}_c$ by central differences on 24 held-out operating points. The co-energy network violates reciprocity by 4.6×10^{-4}—nearly three orders of magnitude below the direct MLP's 0.22—and keeps a positive-definite symmetric part (min eigenvalue $0.36 > 0$, physically admissible incremental inductances). The monotone vector field is exactly symmetric by construction (1.6×10^{-5}) but, as the next section shows, fits nothing.

Table 4: Coverage-kind ablation (flux nRMSE / torque R^2). Hold-outs: openB = phase-B open circuit, tap50 = half-winding series tap, delta = star–delta recombination. Magnitude patterns (48 groups, 4,608 FE samples) and phase patterns (24 groups, 1,152 samples) rescue *different* hold-out connections; the union covers all.

Training pool	openB	tap50	delta
standard only	1.83/−0.62	1.86/0.41	0.27/0.84
+ magnitude (48 groups)	1.38/ 0.29	**0.43**/0.70	0.36/0.84
+ phase (24 groups)	**0.07**/**0.99**	2.11/−0.01	**0.07**/**0.99**
+ both	0.09/0.97	0.43/**0.73**	0.08/0.98

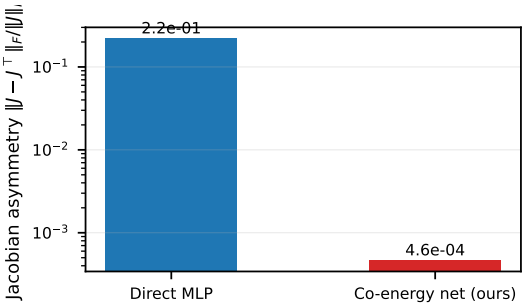


Figure 4: Flux-Jacobian reciprocity residual on held-out operating points (log scale): the co-energy network violates reciprocity nearly three orders of magnitude less than the direct MLP.

5.4. Negative result: monotonicity is the wrong structure

The monotone vector field enforces joint convexity of the underlying scalar in $(\boldsymbol{i}_c, \phi)$. The true co-energy is approximately convex in $\boldsymbol{i}_c$ alone at fixed θ, but its θ-dependence through the PM field and slotting makes the joint Hessian indefinite; the architecture can therefore only fit a compromise, and lands at flux nRMSE 0.28–0.99 and strongly negative torque R^2 at every tier, including in-support (4S LOCO 0.41, single-connection-trained variant 0.48), so the failure is not an artefact of cross-connection sharing. Stronger training (2× epochs, 3× learning rate) does not close the gap either: the failure is structural rather than one of optimisation. The lesson for structural design in magnetic ML: *enforce the physical symmetries the PDE actually has (reciprocity, potential structure),*

Table 5: Jacobian reciprocity and positive-definiteness on held-out points (asym).

Model	$\|J - J^\top\|_F / \|J\|_F$	min eig$(\frac{J+J^\top}{2})$
Direct MLP	0.221 (max 0.363)	0.340
Co-energy (ours)	4.6×10^{-4} ($\max\ 2.2 \times 10^{-3}$)	0.359
Monotone VF	1.6×10^{-5}	0.278

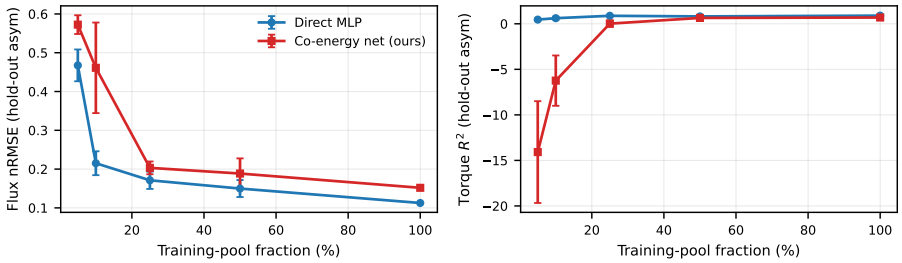


Figure 5: Data-efficiency learning curves on the hold-out connection (three seeds): flux nRMSE (left) and torque R^2 (right) vs fraction of the training pool.

not convenient convexity properties it does not have. Scalar co-energy parameterisations get the former for free.

5.5. Data efficiency and coverage scaling

Scaling the magnitude-pattern family on the unbalanced hold-out (asym, MLP): 0 groups 1.00, 2 groups 0.51, 8 groups 0.22, 24 groups 0.16, 48 groups 0.11—a clean coverage curve that saturates near 24–48 groups, i.e. a few thousand FE samples. On delta the same family leaves the error essentially unchanged (0.27 at zero groups, 0.26 at 24, 0.36 at 48), while the quarter-sized phase family reaches 0.07 (§5.2), which is the coverage-kind argument in its purest form.

Fig. 5 shows subsampling of the full training pool (three seeds, hold-out asym): the MLP reaching nRMSE 0.22 with only 10 % of the data (497 samples) and 0.17 at 25 %; the co-energy net needs ∼25 % of the pool to reach its plateau (0.15–0.19) and its torque channel degrades more sharply in the extreme low-data regime (5 %: $R^2 < 0$), consistent with its gradient-based outputs needing

Table 6: Cross-machine replication on the 6-slot/4-pole SPM. Entries are flux nRMSE / torque R^2 for the direct MLP. The co-energy network was not re-trained for this replication, so the table tests the two sampling principles rather than architecture ranking.

Hold-out	Standard	+ magnitude	+ phase	+ both
tap50	0.620/0.753	**0.296**/**0.821**	0.648/0.788	0.304/**0.881**
openB	1.110/0.753	1.079/0.783	0.082/**0.997**	**0.065**/0.992

more support to anchor the residual head. The practical reading: with adequate pattern coverage, a few hundred FE samples already yield usable zero-shot flux models; the structured head is worth its consistency guarantee once at least a quarter of the pool is available.

5.6. Cross-machine replication

To test whether the two coverage principles are artefacts of one machine, we repeated the core ablations on a second machine with the same solver family: a 6-slot/4-pole SPM (different slot count, winding factor, inductance matrix and harmonic content; 3,084 FE samples in the same dataset families). With a direct MLP and the identical protocol, the pattern repeats: the half-winding tap hold-out improves with *magnitude*-random patterns (flux nRMSE $0.62 \rightarrow 0.30$; phase patterns alone leave it at 0.65), while the open-phase hold-out is rescued by *phase*-random patterns ($1.11 \rightarrow 0.08$; magnitude patterns alone leave it at 1.08). The union pool again serves both (0.30 and 0.065). Which family a connection needs tracks the geometry of its current vectors, not the machine—as the coverage argument requires.

6. Drive-cycle co-design case study

6.1. From surrogate to system losses

Unless stated otherwise, the drive-cycle study uses the direct-MLP surrogate as the reference model: it is the most accurate of §5 on the connections involved, and the co-energy surrogate is used below as an independent cross-check that

brackets the conclusions. The surrogate is queried in a synchronous-rotating frame. For a mode m with connection matrix $\boldsymbol{C}_m$, line-current amplitude I and current angle γ, the phase currents rotate as $i_p(t) = I[\cos(2\theta+\gamma), \cos(2\theta+\gamma-\frac{2\pi}{3}), \cos(2\theta+\gamma+\frac{2\pi}{3})]$. Average torque, copper loss, the iron-loss proxy and the port voltage then follow from one electrical period of surrogate evaluations (12× torque grid, 72× voltage grid), giving loss maps $P^m_{\text{loss}}(T,\omega)$ and voltage envelopes $V^m(I,\omega) \le V_{dc}/\sqrt{3}$ per mode in under a second on GPU. The loss model comprises exact copper loss $\sum_c i_c^2 R_c$ from the coil currents; inverter conduction and switching from declared device parameters ($V_{CE0} = 0.85\,\text{V}$, $r_{CE} = 9\,\text{m}\Omega$, $E_{sw} = 2.7\,\text{mJ}$ per period at 10 kHz); a Bertotti-type [21] iron-loss proxy on the AC coil-flux amplitude (coefficients stated in the released configuration, ±2× swept); and a changeover energy E^c_{sw} per reconfiguration event (5 mJ–100 J swept). The vehicle is a 540 kg light EV (gear 13, wheel 0.25 m, $C_dA = 0.50\,\text{m}^2$, 320 V bus), and the cycle is the official WLTC class-1 trace [22] (JRC data, checksum-verified). The trace spans the machine's full mode space: below ≈57 km/h all four modes are feasible, above it 4S drops out on voltage, and only 2S2P/delta/4P reach the 64.4-km/h cycle maximum. Only 4P spans the whole cycle, but it is current-class-inefficient below 57 km/h (four times the line current of 4S for the same torque), while 4S is the efficient low-speed mode but voltage-infeasible above it—reconfiguration therefore pays off—precisely the industrial setting.

6.2. Scheduling formulation

With per-step mode-conditioned optimal operating points, mode selection is a shortest-path problem over a mode graph with dwell constraints. We solve it by dynamic programming on the state (mode, steps-since-last-switch) with minimum dwell $D = 10\,\text{s}$, comparing: **fixed-4P** (the only single mode covering the whole cycle—the honest no-reconfiguration baseline), **2-mode rule** (4S/4P at a swept speed threshold, the practice of [2]), **pointwise greedy**, and **DP**.

6.3. Results

Table 8 and Fig. 6 summarise the case study.

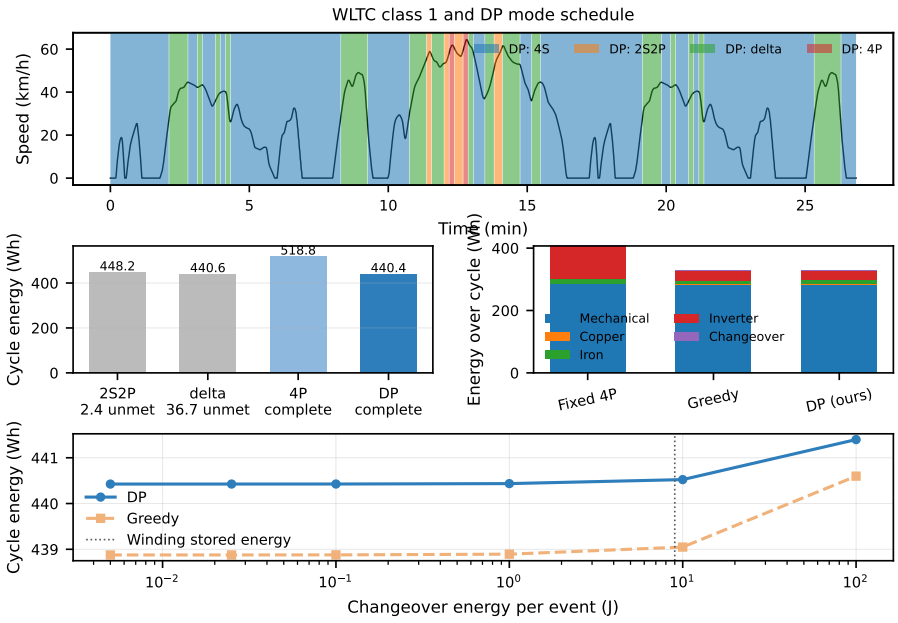


Figure 6: Drive-cycle case study. Top: WLTC class-1 speed trace and the DP mode schedule (colour band). Middle-left: fixed-connection and DP cycle energy; incomplete trajectories are marked by their unmet torque integral. Middle-right: loss decomposition for completed trajectories. Bottom: sensitivity of cycle energy to the changeover energy, swept from 5 mJ to 100 J.

Reconfiguration buys feasibility without a material energy penalty. Fixed 4P—the only single connection that spans the whole speed range—consumes 518.8 Wh (45.4 Wh/km) over the 11.43-km WLTC class-1 trace. The DP schedule consumes 440.4 Wh (38.5 Wh/km), uses 35 changeover events (3.1 per km), and tracks the cycle with no torque deficit. Against 4P this is a 15.1 % reduction, but 4P is a deliberately weak fixed baseline at low speed: it carries four times the line current of 4S for the same torque and is the only fixed mode forced to span the whole cycle. The decision-relevant comparison is therefore with the best *nearly* feasible fixed connection. Fixed delta reaches 440.6 Wh on its incomplete trajectory while missing 36.7 N m s of requested torque; fixed 2S2P reaches 448.2 Wh while missing 2.4 N m s. DP's energy is 0.03 % below delta's partial-trace energy and 1.7 % below 2S2P's. Because

incomplete trajectories are not directly comparable, the robust result is not an energy gain over delta but *feasibility at essentially the same energy*. Other fixed connections are worse: 4S misses 541.4 N m s, the zero-shot 3S1P candidate misses 302.9, and tap variants miss 308–1,332 N m s.

The loss decomposition explains the large gap to fixed 4P: inverter conduction drops from 104.3 to 31.2 Wh because low-speed cruising runs in 4S at a quarter of the line current; iron loss drops from 16.1 to 10.3 Wh; copper loss is essentially unchanged (2.7 vs 3.3 Wh). Part of this gap reflects one inverter rated for the largest current class of any mode. With current-class-matched devices the DP-vs-4P saving is 13.3 %. We retain that number as a baseline sensitivity, not as a headline energy benefit of reconfiguration.

Optimal scheduling matches the unconstrained pointwise optimum at half the switching burden. The per-point greedy oracle (no dwell, no changeover penalty) gives 438.9 Wh with 62 switches; DP reaches within 0.4 % of it with 35 switches. The residual 1.5 Wh (5.4 kJ) equals 200 J per forgone switch—an order of magnitude above any plausible per-event changeover cost (the winding stores ≈ 9 J at typical commutation currents)—so the gap is the price of the 10-s dwell constraint, not of changeover energy. The classical two-mode speed-threshold rule cannot complete the cycle on this powertrain, incurring 419 N m s of unmet braking torque at the mode boundary: 4S is voltage-infeasible above the threshold, while 4P cannot deliver the required low-speed braking currents without the scheduler. Its nominal 264 Wh therefore corresponds to a cycle that is only partially driven and is not comparable. This is the failure mode that motivates schedule-level optimisation over rule-based switching.

The saving is robust across physically meaningful changeover energies. The default per-event cost (25 mJ) sits far below the physics: the winding stores ≈ 9 J of magnetic energy at typical commutation currents, which any real changeover device must dissipate or commutate. Sweeping the per-event energy over four orders of magnitude—5 mJ to 100 J—leaves the DP schedule identical (35 events at every value) and costs at most 1.0 Wh (0.2 %) at the extreme end;

even at 100 J per event, ten times the stored magnetic energy, the schedule still completes the cycle and the added energy is at most 1.0 Wh. Scaling the iron-loss coefficients by 0.5× and 2× moves the DP-vs-4P difference only between 14.7 % and 15.5 %. These sweeps establish robustness of the scheduling decision, not a universal topology-energy benefit.

Device rescaling and surrogate uncertainty. The baseline uses one inverter for all modes, which flatters low-current modes (4S at a quarter of 4P's line current). Re-running with current-class-matched device parameters (r_{CE} halved, V_{CE0} +10 %) changes the DP-vs-4P difference from 15.1 % to 13.3 %. Re-evaluating the same DP schedule with the independent co-energy surrogate gives 21.7 % against 4P. The 6-point spread exceeds the device-rescaling effect and shows that absolute surrogate savings are uncertain; the feasibility gap to incomplete fixed alternatives remains the load-bearing conclusion. The mode histogram is physically interpretable: 4S dominates low-speed urban segments (1,114 s), delta carries mid-speed cruising (416 s), 4P appears only in the highest-speed phase (22 s), and 2S2P is used marginally (59 s).

6.4. Zero-shot modes and connection screening

The drive study above schedules four connections that all appear in the training pools. The point of the connection-invariant formulation, however, is that *any* $\boldsymbol{C}$ can be queried without new FE data or retraining. Two experiments exercise this.

A zero-shot candidate exposes the limit of screening. We add tap50—never present in the drive surrogate's training pool in any form—as a fifth switchable mode. The surrogate assigns it zero seconds, which would suggest that 2S2P/delta dominate it on the cycle. The independent FEMM trajectory check does not support that action: the zero-shot flux correlation is only 0.71 after global scaling, and the torque correlation is −0.21. The flux trend is informative enough to generate a hypothesis, but the torque channel is not reliable enough to reject the connection. We therefore remove the earlier "dominated" conclusion and report this as a negative result for automatic

Table 7: Exploratory zero-shot screening: fixed-connection cycle energy and unmet torque integral (WLTC class 1). Entries marked * are absent from the drive training pool. The tap50 FEMM check shows that these surrogate rankings require targeted FE validation before use as design decisions.

Connection	Energy (Wh)	Unmet torque (N m s)
4S	(215.0)	541.4
2S2P	448.2	2.4
delta	440.6	36.7
4P	518.8	0.0
3S1P*	(258.7)	302.9
tap75*	(105.9)	1,050.4
tap50*	(109.1)	308.2
tap25*	(90.8)	1,331.6

zero-shot screening.

Connection screening at design time. Table 7 evaluates eight candidate connections (including three tap ratios and an unbalanced 3S1P split, all zero-shot for the tap/3S1P entries) as fixed connections over the same cycle. It is a screening experiment, not a validated ranking: the tap50 FEMM check above shows that the torque channel can fail on an unseen synchronous path. The table is useful for forming FE validation priorities in minutes, but a candidate that survives this screen still requires targeted FE checks before a design decision.

7. Discussion and limitations

Design principles for connection benchmarks. The two coverage principles (magnitude families for unbalance, phase families for phase-recombining connections) are motivated by Prop. 2. We expect them to carry over to other machines: pattern families should be chosen by the span of the target connections' current vectors, not by sample count. Multi-phase and fault reconfiguration studies are the natural next testbeds, where the "phase" family is the default recommendation.

Table 8: Drive-cycle case study: WLTC class-1, 540-kg light EV, 320-V bus, 11.43 km, all losses counted, minimum dwell 10 s. Energy in parentheses belongs to an incomplete trajectory and is descriptive only. The comparison column is therefore shown only for completed trajectories.

Strategy	Complete	Energy (Wh)	Wh/km	vs DP	Switches	Unmet torque (N m s)
Fixed 4S	no	(215.0)	—	—	0	541.4
Fixed 2S2P	no	(448.2)	—	—	0	2.4
Fixed delta	no	(440.6)	—	—	0	36.7
Fixed 4P	yes	518.8	45.40	+17.8 %	0	0
2-mode rule	no	(264.0)	—	—	6	419
Pointwise greedy	yes	438.9	38.40	−0.4 %	62	0
DP (ours)	yes	440.4	38.54	0	**35**	0

What structure buys. On the *flux* channel the structured co-energy network improves on the MLP for tap50 and openB but not for asym, and concedes accuracy on tier-1 hold-outs, in exchange for exact potential structure. Its *torque* channel, however, is consistently weaker (R^2 as low as -18.5 on openB, -0.05 on delta): differentiating a learned scalar amplifies residual error, so torque-critical applications should either up-weight the torque loss further or accept the MLP's pointwise torque. In downstream use where consistency matters (optimisation, control, port-level circuit solving) the structured model can never produce non-conservative operating points; the Jacobian evidence (Table 5) quantifies the difference. Our benchmark makes this trade-off explicit rather than hiding it.

Limitations. (i) 2-D magnetostatics with a research-grade solver; the fine-mesh check (Table 1) bounds discretisation uncertainty at a mean 10 % (max 30 %) on flux and 13 % median on torque. This is of the same order as the reported model errors (0.04–0.4 nRMSE), so *absolute* accuracies should be read as accuracy relative to this discretisation. The FEMM cross-check corroborates matrix and waveform trends but also shows 11.7–40.9 % discrepancies and a failed zero-shot torque screen on tap50 (Table 2); it therefore does not establish absolute accuracy. (ii) End effects, temperature, mechanical and changeover transients are excluded; the changeover circuit enters through a per-event energy, swept over four orders of magnitude (5 mJ–100 J). (iii) The drive-cycle

study is a simulation case study on a laboratory-scale machine; absolute savings percentages are machine- and vehicle-specific, and we make no homologation claims. (iv) Iron loss is a Bertotti proxy calibrated to the assumed steel curve, not measured. (v) External validity: the quantitative benchmark and the drive-cycle case study derive from a single machine; the two coverage *principles*, however, are replicated on a second machine (§5.6). All ground truth remains simulation-only, and no measurement validation was performed. A commercial FE cross-check is now included (Table 2), but it exposes both discretisation/model-form differences and a zero-shot torque failure on an untrained tap connection. Replication on a measured machine remains required before the design principles can be treated as general.

8. Conclusion

A winding connection is a linear constraint, not a new machine. Encoding that fact into the learning problem yields one surrogate for all connections, zero-shot transfer governed by current-space coverage, and exact physical consistency from a scalar co-energy head. It also yields a drive-cycle co-design loop that prices reconfiguration with full loss accounting—changeover energy, dwell constraints and system losses. On the WLTC class-1 case study, the optimal schedule completes the cycle at essentially the same energy as the best nearly feasible fixed connection, adds the missing tracking capability, and uses 3.1 changeover events per km. It is within 0.4 % of the unconstrained greedy oracle at half its switching burden; the 15.1 % advantage over fixed 4P is reported only together with that baseline's current-class penalty. The benchmark and its two data-design principles are, we believe, the most durable contributions: they tell the next study, whatever its machine, what data to collect and why. Future work will address geometry-parametric co-energy operators (design-space generalisation), thermal coupling, and hardware validation of the scheduling layer on a changeover test rig.

Reproducibility

All FE data (23,000+ samples across the benchmark and drive-cycle pools), the solver, the training and evaluation code, trained checkpoints, benchmark splits, the official WLTC class-1 trace preprocessing, the second-machine replication scripts, and one-command figure scripts are released at [[repository URL on acceptance]].

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